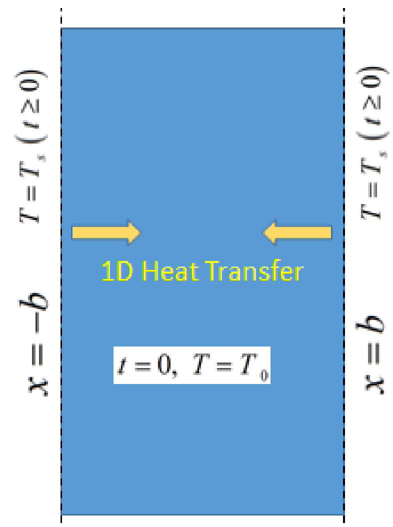
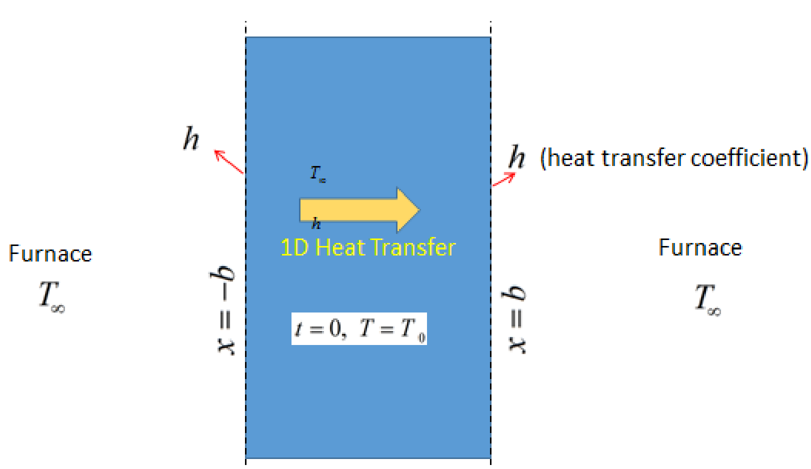
Practice Assignment (Unsteady Heat Transfer)

- A solid slab with initial temperature T_0 is kept between $x = \pm b$, as shown in figure below. Suddenly at $t = 0$ the temperature of the boundaries ($x = \pm b$) has been raised to T_s and maintained for $t > 0$.

Find the expression for temperature profile $T(x,t)$. Consider 1D heat transfer in x direction (y and z directions are infinite). Other details: κ, ρ, C_p are thermal conductivity, density, and specific heat capacity respectively.



- A solid slab with initial temperature T_0 is kept between $x = \pm b$, as shown in figure below. Suddenly at $t = 0$ the slab is placed in a furnace maintained at temperature T_∞ for all $t \geq 0$. Find the expression for temperature profile $T(x,t)$. Consider 1D heat transfer in x direction (y and z directions are infinite). Other details: κ, ρ, C_p are thermal conductivity, density, and specific heat capacity respectively. ***h*** is heat transfer coefficient at two boundaries exposed to furnace.



- A solid material occupying the space from $x = 0$ to ∞ is initially at temperature T_0, as shown in fig (i). At time $t = 0$, the surface at $x = 0$ is suddenly raised to temperature T_1 and maintained at that temperature for $t \geq 0$. (Assume **one dimensional heat transfer in +*x* direction**, thermal diffusivity of material is α).

a) Write down governing equation and initial/boundary condition.

b) Find the time-dependent temperature profiles $T(x, t)$.

c) Show that wall heat flux, $q|_{x=0}$ varies with time (t) as, $q|_{x=0} \sim t^{-1/2}$.

d) Prove that the penetration depth (δ_T) is given by $\delta_T = (\pi \alpha t)^{0.5}$, where t is the real time.

Penetration depth is defined as the location x, where the tangent to temperature profile ($T(x, t)$) for any t at $x = 0$ intersects the line $T = T_0$ as shown below in figure (ii)
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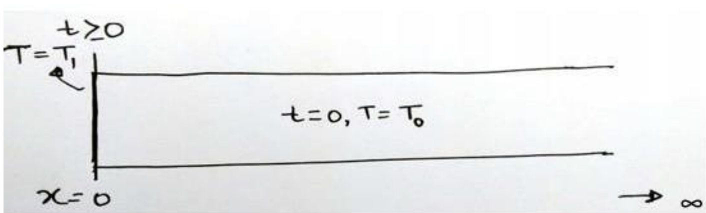

Figure (i)

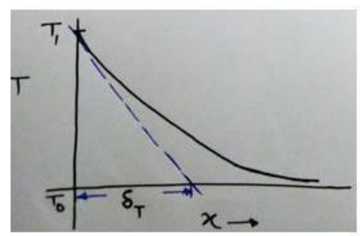

Figure (ii)

$$erf(z) = \frac{2}{\sqrt{\pi}} \int_0^z \exp(-t^2)\,dt, \qquad erfc(z) = 1 - erf(z).$$

$$\frac{d(erf(z))}{dz} = \frac{2}{\sqrt{\pi}} \exp(-z^2)$$